

market with the Google Cloud Platform which looks alluring initially due to its zero costs. It may become more expensive as the projects grow in complexity. But there are some minimum requirements for the setup to ensure a smooth simulation run with no crashes.

Experts suggest Intel Core i5 6th Generation processor or higher as mandatory. Having an AMD equivalent processor works just as well. When it comes to RAM, at least 8GB to 16GB is recommended.

There are many options when it comes to choosing GPUs. Start with the simpler NVIDIA GeForce GTX 960 or higher. Be wary about using AMD GPUs as they are not the default choice for deep learning exercises. And if any problem occurs while configuring the GPU with the system, rebooting manually is not feasible. GPUs are nevertheless great for parallel programming and can be used to analyse datasets other than text and numeric types.

What about operating systems? Windows is always the first choice regardless of version. But across your journeys using Keras, you have to confront the need for greater CPU strengths which can be achieved by linking multiple PCs together. Ubuntu and Linux OS work well in configuring cycle speeds for singular procedures. If you have a laptop that runs on similar operating systems, getting some additional RAM would power up the simulations. It would be ideal to use high end models from brands like Alienware, ASUS, and lineups like Acer Predator or Lenovo Legion.

You might be swayed initially to invest in a GPU given its ability to clock speeds 2 to 3 times faster than a CPU, but it is best to wait out and check for appropriate prices. There are alternatives to the GPUs such as FPGAs and ASIC. Another alternative is to use the CUDA driver that was created by NVIDIA for graphic processing in high level languages. Versions like the i7-7500U can train an average of 120 calculations/second.

Checklist for best practices

1. Start low and start well. Use conventional types like the NVIDIA GTX 1080 (8 GB VRAM) that can bring about 14000 computations per second. Remember that the GPU and CPU must work simultaneously for the best results. Using an i7-7500U will run smoothly with a GTX 1080 GPU for example.
2. Furthermore, GPUs can perform convolutional/CNN or recurrent neural networks/RNN based operations much faster which may warrant their use against TPUs. It depends on what the operation demands and what can be done to optimize the systems better. It can also perform operations